

Product Manual

Kamoer Fluid Tech(Shanghai) Co., Ltd.

Version : A/0

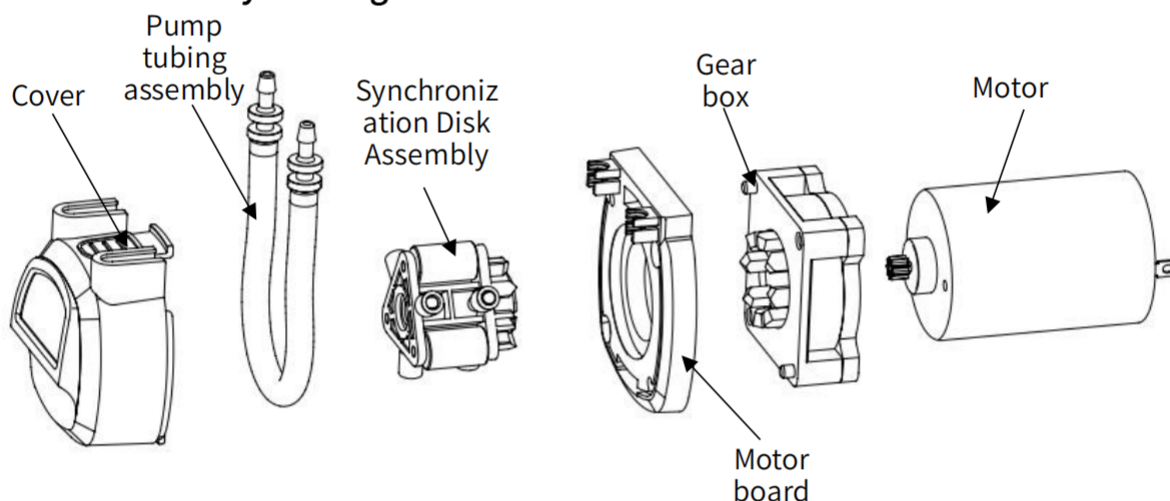
Product name	Micro peristaltic pump
Product model	KHS2
Execution date	2023.07.18
Company name	Kamoer Fluid Tech (Shanghai) Co.,Ltd.

A、Product Overview

1. Product picture



2. Product assembly drawing



3. Model definition

Model	Gearbox	Motor model	Rotor	Cast	Tube number	Installation method
KHS 2	2	SV	3	N	16	null



3.1 Motor model



SW 12V DC Brush
SV 24V DC Brushed



2- SW 12V DC Brush
2- SV 24V DC Brushed



12B 12V DC brushless
24B 24V DC brushless



2- 12B 12V DC brushless
2-24 B 24 V DC brushless

3.2 Rotor

3 rotors

3.3 Tube type

N: Norprene

3.4 Tube number

Tube number	16#	40#
Metric system	3.2mm x 6.4mm	4.0mm x 7.2mm
Imperial	1/8in x 1/4in	5/32in x 9/32in

3.5 Installation method

Empty: standard installation

3.6 Product material

Pump head model	Cover	Output chainring	Gear	Bracket	Connector
KHS2	PA	POM	PTE	PTE	PP

4. Performance characteristics, typical applications

4.1 Performance characteristics

◆ KHS series peristaltic pump is a peristaltic pump driven by 2 motors, 24V/12V motor, which can provide different voltage options

◆ The pump tube is imported Norprene (N) tube with a wall thickness of 1.6mm , other pump tubes can be customized

◆ The synchronous plate of the main working parts adopts upper and lower rollers to ensure that the pump tubes work at the same level

◆ There are two installation methods: bracket fixing and veneer through-board fixing to suit different usage occasions (see the installation diagram for details)

◆ The structural parts of the pump body are made of high-strength synthetic engineering plastics to ensure the long-term reliable operation of the pump

◆ Transmission mode - The transmission mode is gear transmission, the first-level reduction ratio is 1:8, the second-level reduction ratio is 1:64, with high reliability and low noise.



4.2 Typical applications



Water quality analyzer



Nutrient solution ratio



Aquarium water change

Analytical instruments: water quality analyzer, nutrient solution ratio, aquarium water change

5. Product Certification, Intellectual Property

Appearance design



Utility model



RoHS certification



CE certification



◆ Hoses are resistant to liquid media Need to check chemical compatibility or conduct immersion tests, improper hose selection may cause hose damage quickly

◆ The working environment of the product should not exceed 45°C, and the humidity should not exceed 70% (no condensation). The harsh working environment will cause premature damage to the product

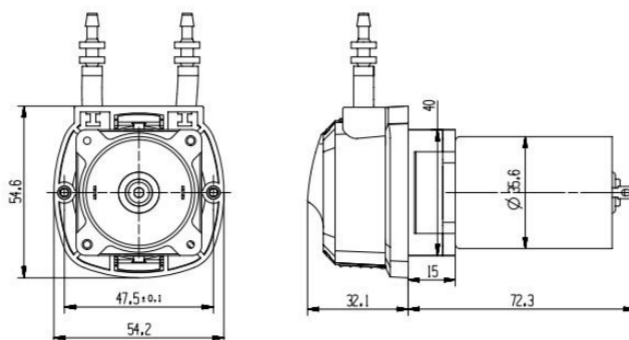
◆ Fluid leakage accidents due to hose ruptures depend on the fluid medium and your specific application conditions

◆ High overloading may result in premature failure of the product

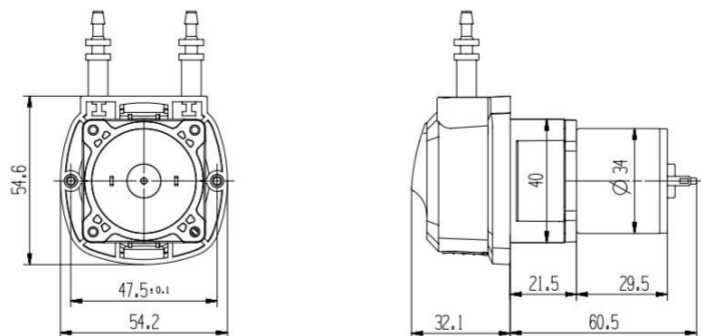
B、Product specification

◆ DC Brush Product Specifications

1. KHS2-2-SV3N16/N40 Product Dimensions



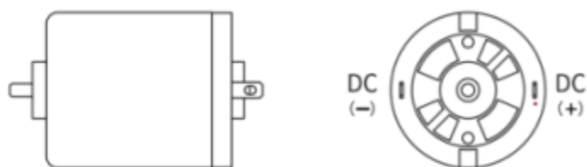
2. KHS-2-SW3N16/N40 Product Dimensions



3. Technical Param

Project		Technical Parameters		
Basic parameters	Motor type	DC brushed motor		
	Flow range	30ml/min-310ml/min; precision ± 10%		
	Gearbox	Level one	Secondary	
	Noise	≤68dB	≤60dB	
	Power	10W	5W	
	Control method	Switch control		
	Weight	About 223g	About 155 g	
	Rated voltage	12V/24V		
	Pump tube life	N : 10 00h		
	Motor life	800h		
	Reference noise value	Environmental noise 35dB , 50cm away from the actual measured first-level noise is less than 68dB , second -level noise is less than 60dB		
Pump flow	Reference flow rate (unit: ml/min)			
	Pump tube code	N16	N40	
	inner diameter x outer diameter	Metric: 3.2mm x 6.4mm Inch: 1 / 8 in x 1 / 4 in	Metric: 4.0mm x 7.2mm Inch: 5/32 in x 9/32 in	
	Pump tube material	N		
	Level one	12V/current : 0.8A	190ml/min	240ml/min
		24V/current: 0.45A	210ml/min	270ml/min
	Secondary	12V/current : 0. 4 A	36ml/min	50ml/min
		24 V/ current : 0.2 A	36ml/min	52ml/min
Supplementary Note		The above parameters are measured with pure water without pressure at room temperature and standard atmospheric pressure at 20°C. Actually, there will be certain errors in the flow rate due to different media, different outlet pressures, and DC motor speed errors. Suitable for intermittent work.The data are for reference only. In addition, it can be customized according to customer needs		

4. Motor control

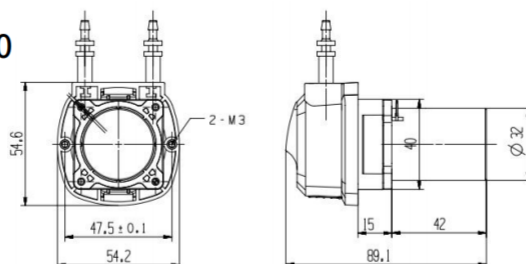


Note: According to the motor wiring diagram, the positive pole of the power supply is connected to the positive pole, and the negative pole is connected to the negative pole.

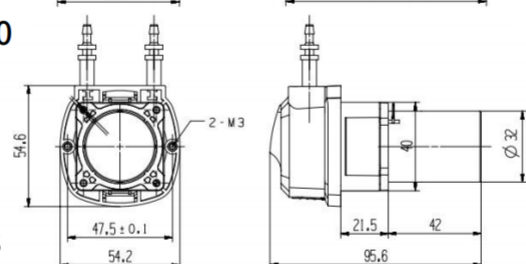


◆ BLDC Product Specifications

1. KHS2-2-12B3N16/N40



2. KHS2-2-24B3N16/N40



3. Technical Parameters

Project		Technical Parameters		
Basic parameters	Motor type	DC brushless motor		
	Flow range	30 ml/min-310ml/min; accuracy±10%		
	Gearbox	level one	Secondary	
	Noise	≤68dB	≤60dB	
	Power	10W	5W	
	Control method	Switching value control/PWM speed regulation		
	Weight	About 164g	About 168g	
	Rated voltage	12V/24V		
	Pump tube life	N : 1000h		
	Motor life	2000h		
	Reference noise value	Environmental noise 35dB , 50cm away from the actual measured first-level noise is less than 68dB , second -level noise is less than 60dB		
Pump flow	Reference flow rate (unit: ml/min)			
	Pump tube code	N16	N40	
	Inner diameter x outer diameter	Metric: 3.2mm x 6.4mm Inch: 1/8 in x 1/4 in	Metric: 4.0mm x 7.2mm Inch: 5/32 in x 9/32 in	
	Pump tube material	N		
	Level one	12V/current: 0.8A	210ml/min	270ml/min
		24V/current: 0.45A	210ml/min	280ml/min
	Secondary	12V /Current : 0.4 A	30ml/min	41ml/min
		24V /current : 0.2 A	30ml/min	41ml/min
Supplementary Note		The above parameters are measured with pure water without pressure at room temperature and standard atmospheric pressure at 20°C. Actually, there will be certain errors in the flow rate due to different media, different outlet pressures, and DC motor speed errors.Suitable for intermittent work. The data are for reference only. In addition, it can be customized according to customer needs		



4. Description of brushless motor wireless correspondence:

◆ The red line is the positive pole of the power supply, the black line is the negative pole of the power supply, the white line is the speed control line, the yellow line is the signal feedback line (for speed measurement), and the green line is the commutation line (suspended to reverse, connect to the negative pole of the power supply for forward rotation).

◆ Speed control port: PWM and 0-5V two speed control methods, speed control range 11%-100% PWM frequency 10KHZ-30KHZ; amplitude: 5V, duty cycle 0-10% motor does not turn, 11%-100% Speed regulation range; 0-5V voltage speed regulation, 0-0.5V motor does not rotate, 0.6V-5V speed regulation range.

◆ The speed feedback of the yellow line is connected to the oscilloscope probe or the host computer. 1 pulse/revolution.

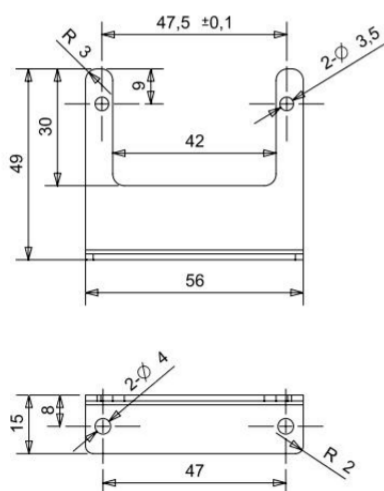
◆ Run at full speed, connect the red and white wires to the positive pole of the power supply, and the black wire to the negative pole.

UL1332 Material: PTFE

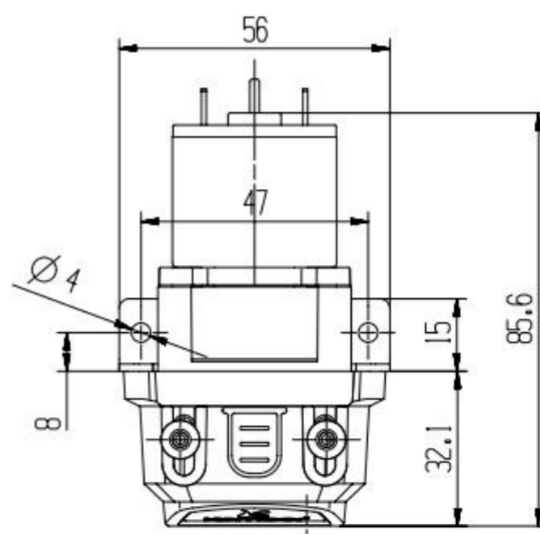
- ◆ FG-24AWG-yellow signal feedback (yellow)
- ◆ Vcc-24AWG-red Positive pole of power supply (red)
- ◆ SP-24AWG-white Speed port (white)
- ◆ GND-24AWG-black Negative pole of power supply (black)
- ◆ F/R-24AWG-green reversing port (green)



C. L board installation method



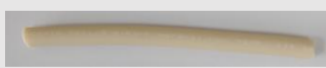
Sheet metal size



L plate installation



D、Adapter hose

Pump tube material	Characteristic	Standards compliant	Tube number
N tube (N) 	Good UV resistance, wide range of chemical resistance, life over 1000 h	RoHS FDA	16#, 40#

E、Optional accessories

Name	Specifications		Function introduction
Straight connector	1/8		3.2mm×6.4mm hose can be connected externally
Straight connector	5/32		4.0mm ×7.2mm hose can be connected externally

Special declaration: The company reserves the right to improve the appearance and technical specifications of the product without prior notice.

